INTEGRATION, the VLSI journal 111 (2026) 102840

Contents lists available at ScienceDirect

Integration, the VLSI Journal

journal homepage: www.elsevier.com/locate/vlsi

TSV testing based on reconfigurable ring oscillator structure

Tian Chen [a,b,c,*], Daping Wu [a,b,c], Yunfei Zhang [a,b,c], Jun Liu [a,b,c,*], Weikun Chen [a,b,c], Xiaohui Yuan [e], Yingchun Lu [d], Huaguo Liang [d]

[a] School of Computing and Information Technology, Hefei University of Technology, Hefei, 230601, China
[b] Intelligent Interconnected Systems Laboratory of Anhui Province, Hefei University of Technology, China
[c] AnHui Province Key Laboratory of Affective Computing and Advanced Intelligent Machine, HeFei University of Technology, China
[d] School of Microelectronics, Hefei University of Technology, Hefei, 230601, China
[e] Computer Vision and Intelligent Systems Lab at the University of North Texas, Denton, TX, 76207, USA

ARTICLE INFO

Keywords:
3D integrated circuits
Built-in self-test
Throughsilicon-via
RLFSR
Ring oscillator

ABSTRACT

Through-Silicon Vias (TSVs) are the key component in stacking three-dimensional chips. It is imperative to detect TSV faults during the manufacturing process. However, the high testing cost and difficulties in testing the entire TSV bonding process using the same structure are open challenges in the manufacturing process. This paper proposes a TSV testing procedure based on a reconfigurable ring oscillator structure. The reconfigurable ring oscillator (RRO) performs fault detection on TSVs Pre-bonding and carries out Mid-bond and Post-bond testing by reusing the technique Pre-bonding. The reconfigurable Linear-feedback shift register(RLFSR) can be employed for circuit testing and generate RRO control signals. Our simulation results of HSPICE substantiate that the proposed structure effectively detects faults at each stage of TSVs. Compared with the state-of-the-art testing techniques, the test data storage volume is reduced by 53.41% and 24.39%, respectively, and the area overhead is reduced by 42.33% and 19.87%, respectively.

1. Introduction

In the development of semiconductor technology, the chip size is shrinking, and the complexity of integrated circuits (ICs) is increasing. The transistor density becomes extremely high. Compared with the traditional two-dimensional integrated circuits (2D ICs), the three-dimensional integrated circuits (3D ICs) based on Through-Silicon Via (TSV) adopt a vertically stacked structure, which has many advantages, including cutting down the overall system size, diminishing the interconnection delay, increasing the operating frequency, and reducing the power consumption. The employment of TSVs significantly enhances the performance [1], which has become the main developing tendency of the next generation of chips.

In 3D integrated circuits, TSVs act as a vertical signal path between diverse dies. In the complex process of manufacturing and bonding, potential faults often occur and lead to the reduction of the performance and reliability of the entire 3D ICs. Accordingly, fault detection of TSV is crucial in the manufacturing process. Existing test methods include external tests based on Automatic Test Equipment (ATE) and Built-in Self-test (BIST). However, the radius of the probes is approximately 35 μm, whereas the diameter of TSVs is about 0.4 μm-4.5 μm, which is much smaller than the size of the probes. In addition, the manufacturing process of TSVs includes thinning operations. After thinning, the thickness of each die layer is about 50 μm, and testing with a probe may cause damage to TSVs [2]. BIST technology integrates test circuits into the chips and realizes self-testing almost without the help of ATE, thus gradually replacing probe testing. Nonetheless, BIST technology requires additional test circuits for each layer of the chips, which adds hardware overhead and introduces latent hazards. It is essential to design a low-cost and reconfigurable 3D chip built-in self-test structure [3].

A variety of BIST techniques have been developed for testing 3D integrated circuits. Sourav Das et al. [1] proposed a pre-bond TSV test method that detects TSV defects by measuring the charging time determined by the equivalent capacitance of a TSV. An open defect shortens the charging time, whereas a leakage defect prolongs it. This method is capable of detecting both open and leakage faults under PVT variations. Mok et al. [4] designed a post-bond TSV test structure consisting of a supply-driven floating-control circuit, which compares the output voltage with the reference voltage of a fault-free TSV to determine whether the TSV within a bridge is defective, thereby improving bridge-fault detection capability. Liu et al. [5] proposed a pre-bond TSV test system based on a self-biased reference current source, which charges and discharges the TSV with a constant current. Faults are detected

* Corresponding authors.
E-mail addresses: ct@hfut.edu.cn (T. Chen), liujun@hfut.edu.cn (J. Liu).

https://doi.org/10.1016/j.vlsi.2026.102840
Received 21 May 2025; Received in revised form 6 June 2026; Accepted 31 July 2026
Available online 7 August 2026

by comparing the charging/discharging time with the reference value of a healthy TSV. When only one type of defect occurs, the system distinguishes between open and leakage faults through the charging time. When concurrent faults occur, it identifies them through the discharging time.

Alternatively, solutions based on ring oscillators were developed for TSV testing. Ring oscillators (ROs) impose minimal area overhead, and the principle is to exploit the relationship between the oscillation period and the TSV capacitance. High-precision defect detection is achieved by comparing the oscillation period of a defective TSV with that of a fault-free one. Based on TSV-induced variations in signal propagation delay, Yi et al. [6] proposed a pulse-shrinking pre-bond TSV testing method that converts rising/falling times into pulse widths, further compresses them into digital codes, and compares these codes with those of a healthy TSV. This method offers high resolution and can detect small-scale defects such as weak opens and weak leakages. The RO structure is enhanced to extend the oscillation period and improve detection accuracy [3,7]. Hao et al. [8] proposed a pre-bond TSV test method using RO-based structures to reduce misjudgment when leakage and open defects occur concurrently. Papadopoulo et al. [9] designed a post-bond test structure using an RO, multiplexers, and demultiplexers, while also considering PVT variations.

In pre-bond TSV testing, one end of the TSV is exposed for test access, whereas the other is buried in the substrate and is inaccessible [10]. RO loops therefore serve as highly effective pre-bond TSV test structures [3,6,8]. However, in post-bonding, the two ends of a TSV are connected to different metal layers across stacked dies and become fully embedded. As a result, pre-bond RO-based test structures cannot be directly reused for post-bond testing, and dedicated on-chip test architectures are desired [11]. For instance, Yu et al. [7] introduced an RO-based structure for post-bond TSV test. Nevertheless, none of the above methods simultaneously detect and distinguish resistive open defects, leakage defects, and their concurrent occurrence with very low area overhead, nor can they test both pre-bond and post-bond TSVs using the same test structure. Additional circuitry is thus required, inevitably introducing extra area overhead.

In response to the above problems, this paper introduces a reconfigurable RO structure based on Reconfigurable Linear-feedback shift registers (RLFSR). Initially, the structure configures existing logic gates in the circuit into NOT gates to form RRO by control vectors generated by RLFSR. And then tests the TSVs Pre-bonding, Mid-bonding, and post-bonding to achieve the reduction of area overhead. The main contributions of this paper include the following three aspects:

1. The proposed structure can be subjected to pre-bond TSV test, mid-bond test, and post-bond test.
2. The structure of RRO is configured by existing logic gates in the circuit, without additional area overhead.
3. The control signal for configuring RO is generated by RLFSR to further diminish area overhead.

In the remaining sections of this paper, we have the following parts: Section 2 introduces the electrical structure and fault model of TSVs. Section 3 provides the specific test structures and test principles. Section 4 presents the comparison of the experimental results of the proposed method and the state-of-the-art solutions. Section 5 concludes this paper with a summary of our work.

2. TSV fault model

TSV faults occur at various stages of manufacturing, affecting TSV performance, connectivity, and overall chip reliability. Figs. 1 and 2 show the electrical failure models in pre-bonding, mid-bonding, and post-bonding, respectively. An electrical model of TSVs is generally established using resistors R, inductors L, and capacitors C. The values are $R_{TSV} \approx 0.06\Omega$, $L_{TSV} \approx 0.27$nH, and $C_{TSV} \approx 0.06$pF [7]. Since the equivalent parameter of the inductor seems very small and has an extremely weak impact on the circuit, it is usually negligible. TSV is established as an RC model in this article.

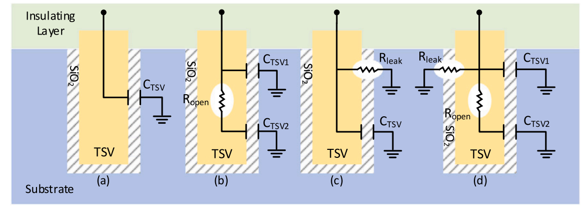

Fig. 1. TSV fault model in pre-bonding.

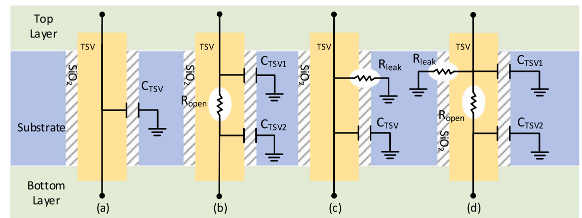

Fig. 2. TSV fault model in mid-bonding and post-bonding.

Fig. 1 shows the failure model of pre-bond TSV. Fig. 1(a) presents the electrical model of a fault-free TSV. Its equivalent resistance is $60m\,\Omega$ which is usually ignored [7]. Therefore, the fault-free TSV is modeled as an equivalent capacitance C_{TSV}. Its capacitance value is expressed as

$$C_{TSV} = \frac{2\pi\varepsilon h}{\ln\left(\frac{d+2t_{ox}}{d}\right)} + \frac{\pi\varepsilon d^2}{4t_{ox}} \tag{1}$$

where h is the length of the TSV in the substrate, d represents the diameter of the TSV, t_{ox} is the insulating layer Thickness, ε is the dielectric constant of the insulating layer [8].

Fig. 1(b)illustrates a TSV with an open fault. During the TSV manufacturing process, imperfections result in holes because the TSV is not completely filled, which may pose a threat to the smooth transmission of signals. The void causes TSV resistive open faults, and the open-circuit resistance R_{open} increases as the void expands. The capacitance of the TSV is divided into upper and lower parts by an open fault

$$C_{TSV1} = \frac{x}{h} C_{TSV} \tag{2}$$

$$C_{TSV2} = \left(1 - \frac{x}{h}\right) C_{TSV} \tag{3}$$

where x means the location of the micropore. When the void reaches its limits, the upper and lower capacitors are completely isolated, causing the TSV to break. At the same time, only the upper half of the TSV remains:

Fig. 1(c) presents the TSV model with a leakage fault. The oxide gap in the insulating layer gives rise to a short circuit between the TSV and the substrate, causing a leakage path. The leakage current flows to the substrate through the leakage resistor R_{leak}. The larger the leakage pore in the insulating layer, the smaller the R_{leak}, and the more serious the fault.

Fig. 1(d) shows the equivalent electrical model of a TSV that simultaneously exhibits an open defect and a leakage defect in the pre-bond stage, a.k.a. composite faults. This model contains both the open resistance R_{open} and the leakage resistance R_{leak}, and the two types of defects simultaneously affect the oscillation period.

In addition, faults during the bonding process may also lead to a decrease in TSV connection quality. Different from pre-bonding, both ends of the mid-bond TSV and post-bond TSV are completely embedded in the circuit. The equivalent model is depicted in Fig. 2. Fig. 2(a) shows a fault-free TSV. Since TSVs are made of highly conductive metal whose resistance R_{TSV} is very small, it is equivalent to a capacitor C_{TSV} [7]. In Fig. 2(b), there is a TSV with an open fault. The open-circuit resistance is labeled as R_{open}, C_{TSV1}, and C_{TSV2} representing the capacitance of the upper and lower parts, which are shown in Eqs. (1) and (2). Fig. 2(c) shows a TSV with a leakage fault. Compared with a fault-free TSV, the leakage fault produces a leakage resistance R_{leak}. In Fig. 2(d), the electrical model of a TSV that simultaneously contains an open defect and a leakage defect in post-bonding is shown. The post-bond oscillation period is also jointly affected by these two types of defects.

3. Reconfigurable RO structure based on RLFSR

This article takes a 3D chip containing three layers of circuits as an example and designs the test structure shown in Fig. 3, which follows the IEEE1500 standard [12]. The boundary register (WBR) in the IP core shell is connected by a unit (WBC) composed of multiple D flip-flops and multiplexers.

During testing, multiple WBCs are connected to form a structure similar to a scan chain. Each die in the structure includes an Reconfigurable LFSR module (RLFSR), a circuit under test module (CUT), a reconfigurable RO module (RRO), a test response analyzer (MISR), a counter module, and a redundant TSV. Among them, as the widely applied BIST test pattern generator in chip testing, LFSR [13] has the superiority of low hardware overhead, simple design, and high data test data compression rate. The RLFSR module includes D flip-flops, multiplexers(Sel, Opt, Dec), XOR gate and seed registers(SEED). RLFSR's seed set is encoded by a test vector set, which is stored in the ROM of the chip. During testing, RLFSR expands the seed set into the corresponding test vector set. This process is called reseeding. The RRO module includes WBR and logic gates. TG1, TG2, TG3, and TG4 are transmission gates. To reduce hardware overhead, the vector set generated by the RLFSR module is used both as the test vector of the circuit test (Reconfigurable Circuit Test) module and as the control signal of the RRO module. The counter module is used to calculate the oscillation period T. By counting the pulses within a specified time t, the oscillation period can be calculated as $T = t/c$.

3.1. RLFSR-based reconfigurable circuit test module (RCT)

The circuit test module consists of three parts: an RLFSR module, a circuit module under test, and a test response analyzer. The RLFSR serves as a test pattern generator which is used for the reseeding operation of test data. Fig. 4 illustrates a detailed structural diagram of the circuit test on the underlying die. Among them, DFF refers to D flip-flop and Opt, Sel, and Dec serve as multiplexers. The non-lower layer Sel and Dec are used to select whether the data comes from this layer or other layers of RLFSR. The SEED register is used to transmit the seed set. The test vector generated by the RLFSR is determined only by the seed and characteristic polynomials.

The RLFSR of each layer is a reconfigurable LFSR module, which can be used for three stages of testing: pre-bond, mid-bond, and post-bond. The size of the LFSR is related to the maximum number of specified bit in the test vectors. Mid-bond TSV testing is the test of the stacked circuit for each layer of the die stack. The dies need to be tested after each bonding to ensure that dies damaged during the bonding process will not participate in the next bonding. Post-bond TSV testing is the test that needs to be performed after all the wafers are stacked to form an integral circuit. The Post-bond testing and Mid-bond testing are similar. For bonded partial chips or complete chips, the scale of the circuit and test vectors increased accordingly. And the number of specified bits in the vector also increased, which directly determines the size of the required RLFSR. Because of that, the length of the seed is longer than the number of stages of the RLFSR of the current layer circuit. Therefore, it is imperative to form a higher-degree LFSR circuit from part of the flip-flops in the other layers' RLFSR and the flip-

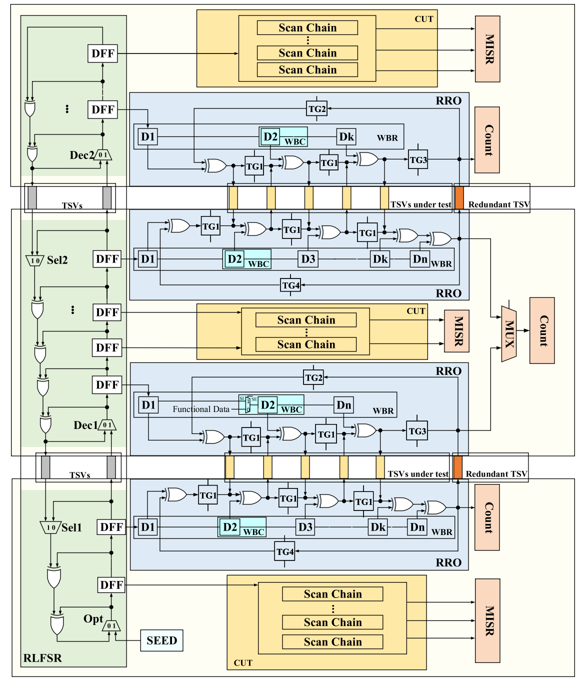

Fig. 3. Overall test structure.

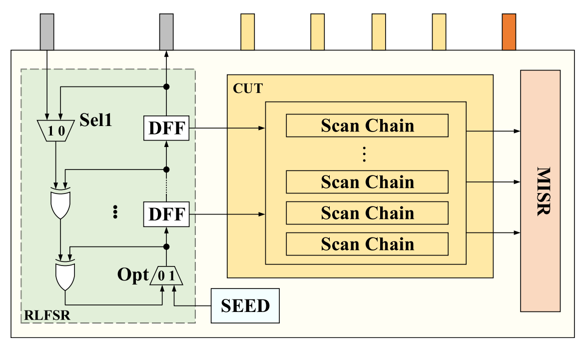

Fig. 4. Underlying circuit.

flops in the bottom circuit. Since we merely utilize the existing RLFSR modules on other layers to construct a larger LFSR, no additional area overhead is introduced. Setting Dec to 1 indicates that the underlying RLFSR module needs to be reused. Setting Sel1 to 1 indicates that the feedback line accepts data from the upper layer RLFSR. The conditions for decompression seeds are met by increasing the degree of RLFSR.

3.2. TSV test module based on reconfigurable ring oscillator (RRO)

The RO is a circuit structure based on the principle of positive feedback and consists of an odd number of inverters [14]. As TSVs under test increase, the ring oscillator's area overhead also expands. To reduce area overhead, the existing logic gates in the die circuit can be fully reused and configured as inverters to form an oscillating ring, as shown in Fig. 5. In this structure, one input value of the XOR gate is fixed to 1 through a D flip-flop, so that its output is opposite to the other input, thereby realizing the function of an inverter. Similarly, logic gates such as NAND, NOR, and XOR gates can also be configured as inverters with fixed inputs of D flip-flops. However, configuring a large number of logic gates requires a large number of control signals. To minimize hardware overhead, our proposed structure reuses the RLFSR

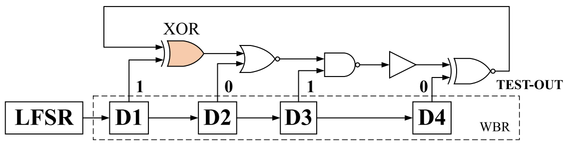

Fig. 5. Oscillating ring structure.

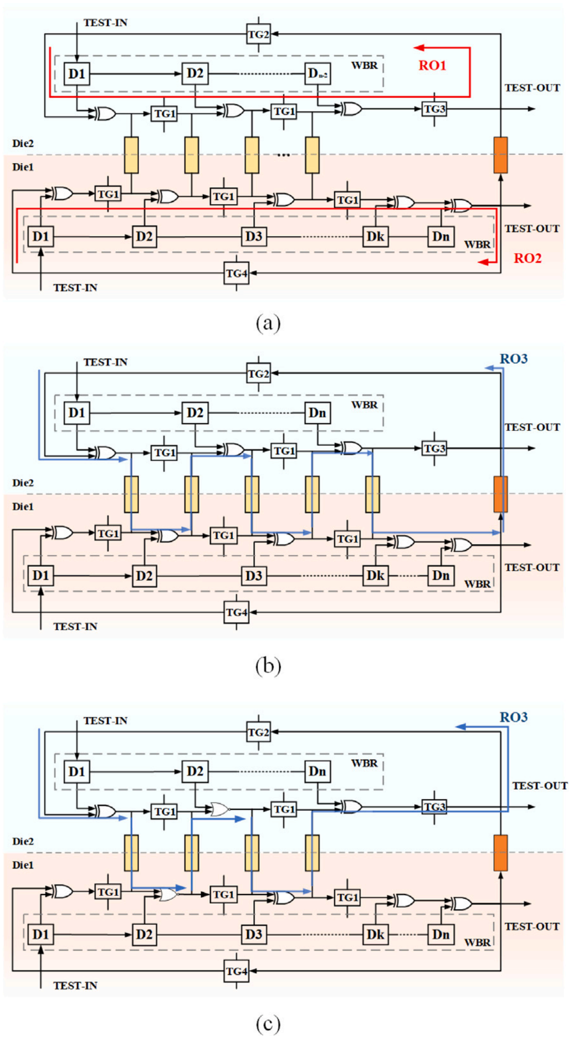

Fig. 6. Reconfigurable RO module. (a)pre-bond test signal path. (b) post-bond test signal path(When TSV is odd). (c) post-bond test signal path (When TSV is even).

module and uses the test vectors generated by it as control signals to configure the configurable logic gates in the circuit.

The working principle of RLFSR is the shift operation and the feedback loop. At each clock cycle, RLFSR bits shift one bit to the right, and the rightmost bit shifts into the scan chain. During the test application process, the RLFSR expands the seed set stored in the ROM into a test vector and moves it into WBR for configuring RRO, as shown in 1010 in Fig. 5. Each test vector is longer than the test seed, and the number of bits that store the seed is much lower than the number of test vectors, so the purpose of compressing test data and reducing the test storage can be achieved.

The control signals used to configure the logic gates can be generated by the LFSR through a reseeding scheme. To enable the LFSR to generate the required control signals on chip, the corresponding LFSR seeds are first calculated and stored in ROM. During testing, a seed is shifted into the LFSR as the initial state, and after several clock cycles, the required deterministic output sequence can be obtained as the control signal for configuring the logic gates. Since the number of bits in an LFSR seed is much smaller than that of the original control vector, the LFSR reseeding method helps reduce the total amount of on-chip storage required for the control signals. For a 3D LFSR structure, better vector compression efficiency and a higher probability of successful seed encoding can be achieved by cascading and reorganizing LFSRs across multiple layers [15].

The number of TSVs determines the different test structures, and Fig. 6 shows the test structures for different numbers of TSVs. Fig. 6 presents the specific structure of the RO module after bonding two layers of dies. The scan chain is composed of the WBC unit in the boundary register. The control signal generated by the RLFSR is input through TEST-IN. TEST-OUT is the signal of the RO output used to calculate the oscillation period of the RO ring. The one TSV on the right is an added redundant TSV; the remaining TSVs are the TSVs under test. The effectiveness of the proposed method is based on the premise that the redundant TSV and the auxiliary test structures have been pre-tested and confirmed to be fault-free. If this premise is not satisfied, the redundant TSV path should be replaced or the test structure should be recalibrated before the formal test is performed. Among them, TG1, TG2, TG3, and TG4 refer to the transmission gates used to control the disconnection and conduction of the circuit. Both Die1 and Die2 include partial RO structures, respectively RO1 and RO2 shown in Fig. 6(a).

During the process of pre-bond TSV testing, TG1, TG2, TG3, and TG4 are all in the on state and forming a loop, such as the red paths RO1 and RO2 in Fig. 6(a), realizing the RO connected end-to-end. At this time, the control signal generated by RLFSR is moved into the scan chain WBR, composed of WBC through TEST-IN, to control the logic gates in the circuit, forming a logic NOT gate structure, and constructing an RO that contains all the required TSV tests. The pulse signal oscillates cyclically in the ring oscillator. After at least two complete cycles, the final output is sent to the counter module via TEST-OUT to calculate the oscillation period. When the TSV has a resistive open fault, the equivalent capacitance of the TSV decreases, and the RC time constant of the RO decreases, resulting in a shortened oscillation period. The number of TSVs does not affect the signal path in pre-bonding, so the signal paths for odd and even numbers of TSVs are the same. When the TSV has a leakage fault, the charge leaks to the substrate through the leakage path, and the equivalent capacitance of TSV increases, causing the RC time constant of the RO to increase, which leads to a longer oscillation period [3]. This method can detect TSV resistive open faults and leakage faults in pre-bonding while determining the type of fault.

In the process of mid-bond TSV testing and post-bond TSV testing, keeping TG1, TG3, and TG4 disconnected and TG2 turned on to form the path RO3 (in blue color) in Fig. 6(b) to ensure that the pulse signal can be transmitted between the two dies through the TSV. Additionally, when the number of TSVs is even, keeping TG1, TG3 disconnected and TG2, TG4 on to form the path RO3 (in blue color) in Fig. 6(c) to ensure that the pulse signal can be transmitted between the two dies through the TSV. The control signal generated by the RLFSR is moved into the scan chain composed of WBC, and the logic gates in the control circuit form a logic NOT gate to construct RO that contains all the required TSV tests. In addition, during the test process, when the number of TSVs is an even number, the pulse signal passes through the TSVs in sequence, and then returns through the feedback line where TG2 is located, entering the next oscillation cycle; when the number of TSVs to be tested is odd, the pulse signal first passes through the TSV that needs to be tested, and finally passes through the redundant TSV, then returns through the feedback line where TG2 is located to oscillate in the next cycle. In post-bonding, whether it is an open fault or a leakage fault, the equivalent capacitance of TSV increases. This causes the RC time constant of the oscillator to increase and the oscillation period

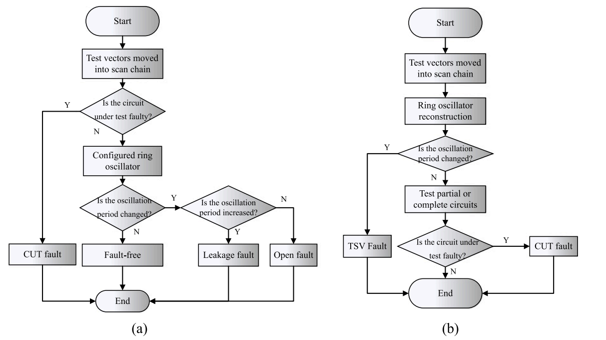

Fig. 7. Test flow chart. (a) pre-bond testing flow chart. (b) mid-bond and post-bond.

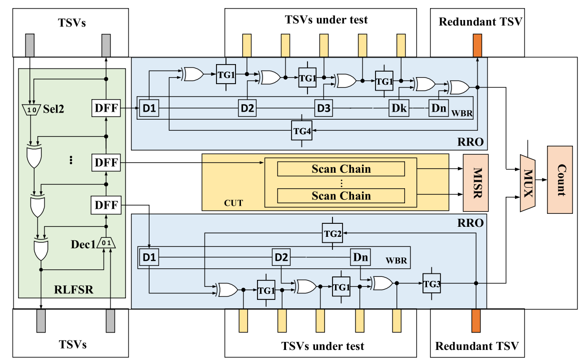

Fig. 8. Single-layer test structure.

to extend. The ring oscillator, therefore, cannot determine the type of fault.

In pre-bond testing, when an open defect and a leakage defect occur simultaneously either within the same TSV or separately in different TSVs, their effects on the oscillation period may cancel each other out. In such cases, the overall oscillation period may appear identical to the fault-free value, making it impossible to detect whether a TSV is faulty in the pre-bond stage. At this point, post-bond testing is required. In post-bonding, both defects have the same influence on the oscillation period and will increase it. Therefore, the ring oscillator can detect TSVs that were missed in pre-bond testing due to the cancellation of the two defects.

In practical circuits, transmission gates are not ideal switches. When a transmission gate is turned on, it introduces on-resistance and parasitic capacitance into the signal path. Therefore, for a test path containing q turned-on transmission gates, the equivalent oscillation period can be approximately expressed as

$$T_{\mathrm{RRO}} \approx T_{\mathrm{logic}} + K\left(R_{\mathrm{drv}} + qR_{\mathrm{on}}\right)\left(C_0 + C_{\mathrm{TSV}} + qC_{\mathrm{TG}}\right) \tag{4}$$

where T_{logic} denotes the intrinsic delay of the logic gates, R_{drv} denotes the equivalent driving resistance, C_0 denotes the equivalent parasitic capacitance of the interconnect, C_{TSV} denotes the equivalent capacitance of the TSV, C_{TG} denotes the equivalent parasitic capacitance introduced by a single transmission gate, and K is a proportionality coefficient related to the number of stages and the structure of the ring oscillator. When a small capacitance variation ΔC_{TSV} occurs in the TSV, the corresponding oscillation-period variation can be further expressed as

$$\Delta T_{\mathrm{RRO}} \approx K\left(R_{\mathrm{drv}} + qR_{\mathrm{on}}\right)\Delta C_{\mathrm{TSV}} \tag{5}$$

Therefore, the on-resistance of the transmission gates mainly changes the sensitivity coefficient between the TSV capacitance variation and the oscillation-period variation, while the parasitic capacitance introduced by the transmission gates mainly changes the baseline oscillation period in the fault-free state. Since the proposed method uses the oscillation period of a fault-free TSV under the same configuration as the reference, the fixed delay introduced by the transmission gates can be calibrated during the reference comparison process. Its main influence on the measurement resolution comes from process variations and mismatch in R_{on} and parasitic capacitance, which increases the required detection margin.

3.3. Overall testing process

Fig. 7 illustrates the overall testing process, which is divided into pre-bond testing, mid-bond testing, and post-bond testing. In the pre-bond testing, the circuit needs to be tested to ensure that there are no problems with the scan chain and circuit. The TSV testing can only be performed if there are no problems with the scan chain configured with the ring oscillator. During mid-bond testing and post-bond testing, the first step is the TSV test. Only by ensuring that there are no problems with each TSV can the circuit test signal be transmitted.

The complete structure of pre-bond testing is shown in Fig. 8, which consists of a RLFSR module, a circuit module under test, and a reconfigurable RO module. The test process is to conduct a circuit test followed by a TSV test. The steps of circuit testing are as follows: firstly, the seed set is sent to the RLFSR. Then the RLFSR decompresses the seed into the corresponding test vector. Moving the generated test vector into the scan chain and then into the circuit under test. Eventually, capturing the test response into the scan chain and moving it into the test response analyzer for result analysis. During the TSV test, initially, TG1, TG2, TG3, and TG4 are turned on. The test vectors are generated by the RLFSR into the WBR as a control signal. The existing logic gates in the control circuit function as logic NOT gates at this time to form an RO to test the TSV, and finally, the oscillation period is calculated by a counter to determine if there is a faulty TSV.

During mid-bond testing and post-bond testing, TSV testing needs to be performed first, and then circuit testing is performed. RO modules between adjacent dies are combined into a complete RO through TSV. The RLFSR of adjacent dies moves control signals into the WBC of this layer to control the logic gates. During the test, TG1, TG3, and TG4 are disconnected, and TG2 is turned on so that the pulse signal passes through all TSVs and returns through the feedback line where TG2 is located. Eventually, the test response analyzer examines the stable oscillation period and determines whether there is a faulty TSV. After the TSV test is completed to ensure that there are no faults, we should test the bonded part or the whole circuit. Setting both Sel1 and Sel2 to 1, the upper RLFSR could form a larger LFSR by reusing the underlying RLFSR module. The data exchange between layers is completed through TSV, and the seed is input into the overall LFSR to generate a test vector to test the complete circuit.

4. Experiments and results

To evaluate the effectiveness of the structure, the TSV test part is simulated using the HSPICE simulation tool under the Nangate 45 nm process library. For a TSV with a 5μm diameter and a 50μm length, its capacitance is 60fF [7]. The experiment is conducted on three-dimensional chips with a three-layer structure, and the LFSR reseeding tool is adopted from [16]. The experiment selects four basic devices: the NOR gate and the XOR gate, as examples. These two basic devices are configured as inverters through the test vectors generated by the RLFSR, with an additional inverter, and the five-stage oscillation ring is formed. For different degrees of open faults and leakage faults, simulations are performed before TSV bonding and after TSV bonding to observe the oscillation period.

Fig. 9 shows the waveform diagram of configuring an oscillation ring. According to the circuit in Fig. 6(c), the experiment was conducted by sequentially configuring XOR, NOR, NOR, XOR, and XOR as invert-

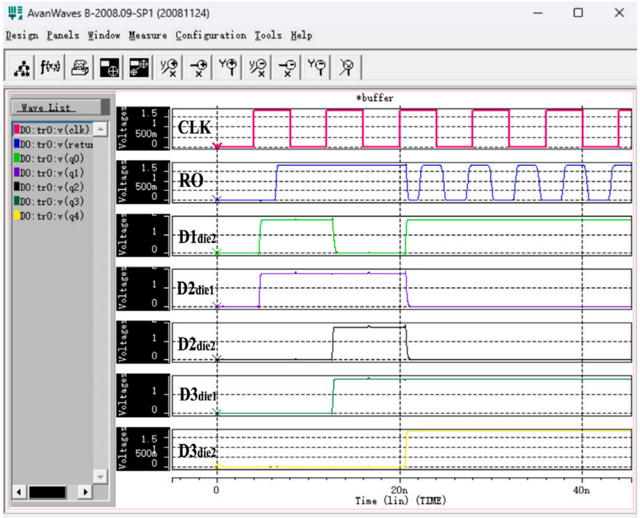

Fig. 9. Oscillating ring configuration process waveform.

ers. Based on the circuit logic, one input of each XOR, NOR, NOR, XOR, and XOR is set to 1, 0, 0, 1, and 1, respectively. The RLFSR generates the configuration sequence [1, 0, 1] for die1 and the configuration sequence [1, 0, 1] for die2 in Fig. 6(c), which are shifted into the D flip-flops connected to the configuration inputs of the XOR and NOR gates. Therefore, the values stored in the D1, D2, and D3 flip-flops of die1 in Fig. 6(c) are [1, 0, 1], and the values stored in the D1, D2, and D3 flip-flops of die2 are [1, 0, 1]. In Fig. 9, the red waveform in the first row represents the RLFSR clock signal (CLK), the blue waveform in the second row is the output of the oscillation ring (RO), and the waveforms from the fourth to the seventh rows are the outputs of the D flip-flops, which are used to configure the XOR, NOR, NOR, XOR, and XOR gates as inverters. The waveform labeled Dndien represents the output of the D flip-flop n in die n in Fig. 6(c). It can be observed that the first three clock cycles are the configuration period, during which the XOR and NOR gates are configured as inverters to form the oscillation ring. In the third cycle, the circuit enters the testing phase, and the outputs of the D flip-flops remain unchanged until the test is completed. In Figs. 10 and 11, “4–2” indicates that there are a total of 4 TSVs, among which 2 TSVs are faulty, and so on.

Fig. 10 depicts the relationship between the TSV open fault and the oscillation period in pre-bonding and post-bonding. It is assumed that the open fault occurs in the middle of TSV, that is, x=1/2h, where x represents the location of the micropore, and h is the length of the TSV in the substrate. Fig. 10(a) and 10(b) reveal the experimental results in pre-bonding and post-bonding when faulty TSVs remain unchanged, and the total number of TSVs continues to increase. According to the line chart, when the number of TSVs with open faults is constant, the oscillation period lengthens as the total number of TSVs increases. Because more TSVs need to be charged at the moment, Fig. 10(c) and 10(d) show the impact of increasing faulty TSVs when the total number of TSVs is held constant, for the pre-bonding and post-bonding phases, respectively. The experimental results show that with the increase of faulty TSVs, the change in speed of the oscillation period becomes more obvious with the severity of the open fault. This is because when the degree of open fault of each TSV and the total number of TSVs remain unchanged, the increase of faulty TSVs leads to a multiple increase in the charging time of the whole TSV.

According to Fig. 10(a) and 10(c), as the open fault increases from $0\,\Omega$(fault-free) to $10k\,\Omega$ (a strong open fault), the oscillation period of the circuit gradually decreases in pre-bonding. The reason is that the open resistance separates the TSV into two parts in pre-bonding. The more significant the open fault is, the greater the equivalent open resistance is, and the more obvious the isolation of TSV is, thus reducing the time to charge the TSV, resulting in a reduction in the oscillation cycle of the entire oscillation ring.

Fig. 10(b) and 10(d) depict that the oscillation period of post-bond TSV gradually increases as the severity of the open fault increases. The reason is that in post-bonding, the open resistance becomes larger because of the TSV located in the oscillation ring, the time to charge two parts of the TSV increases, which leads to a reduction of the oscillation frequency of the oscillation ring, and the oscillation period increases.

Fig. 11 depicts the relationship between leakage TSV fault and oscillation period. The x-axis presents the value of leakage fault resistance R_{leak}. The smaller the value, the more severe the leakage fault is. The electrical models of leakage TSV failures before and after bonding are similar, so the experimental results are consistent. Experimental results demonstrate that as the leakage resistance grows, the oscillation period of the circuit gradually decreases. Once a leakage failure occurs, a short circuit appears between the TSV and the substrate, which forms a voltage-dividing structure and gives rise to prolonged charging time. If the leakage failure becomes more serious, the equivalent leakage resistance becomes smaller, which means that the speed of the current flowing through the leakage resistor to the substrate slows down, thus increasing the oscillation cycle. As the number of TSVs continues to increase, the oscillation period also gradually rises. The reason is that the number of TSVs that need to be charged is also increasing, and the charging time is correspondingly extended.

Fig. 12(a) illustrates the results showing the impact of open defects occurring at different positions on the oscillation period. Since the maximum open resistance used in our experiments is $10\ \text{k}\Omega$, we fix the open defect at $10\ \text{k}\Omega$ for evaluation. In the figure, the horizontal axis represents the oscillation period, and the vertical axis denotes the fault location. When h=1/3, it indicates that the defect is located at one-third of the total TSV height. “4–1 (Pre-bond)” and “4–1 (Post-bond)” represent the oscillation period when one out of four TSVs contains an open defect during pre-bond or post-bond testing, and “Pre-bond fault-free period” and “Post-bond fault-free period” represent the oscillation period when all four TSVs are fault-free. As shown in the Fig. 12(a), during pre-bond testing, the oscillation period increases as the open defect moves closer to the bottom of the TSV, but it always remains smaller than the fault-free oscillation period. This is because, in the pre-bond stage, a resistive-open defect electrically divides the TSV into two segments. The closer the open defect is to the bottom of the TSV, the larger the capacitance of the upper segment becomes, which increases the charging time of the TSV and consequently enlarges the oscillation period of the entire ring oscillator. In the post-bond test, the oscillation period decreases as the open defect approaches the bottom, but it always remains larger than the fault-free period. The reason is that, in the post-bond stage, a resistive-open defect divides the TSV into two segments. The closer the open defect is to the bottom of the TSV, the smaller the capacitance of the lower segment becomes. Since the magnitude of the open resistance is fixed, the charging current is also fixed. Therefore, a smaller capacitance leads to a shorter charging time, resulting in a reduced oscillation period.

In both pre-bond and post-bond tests, when the open fault is located near the bottom of the TSV(h=1), the effective TSV capacitance observed from the test terminal is almost not cut off. Therefore, the equivalent capacitance of the TSV is nearly the same as that in the fault-free state, and the oscillation period of the RRO is also close to the fault-free reference value. As a result, such a boundary-position fault is difficult to distinguish based only on the oscillation-period variation. However, when the fault is located at the very bottom of the TSV, it has only a negligible influence on the charging/discharging path of the test node and the equivalent RC delay. Therefore, under the equivalent capacitance/delay-based test model adopted in this work, this case may approximately exhibit a fault-free response, rather than a typical open fault that can be reliably identified through period variation. A similar

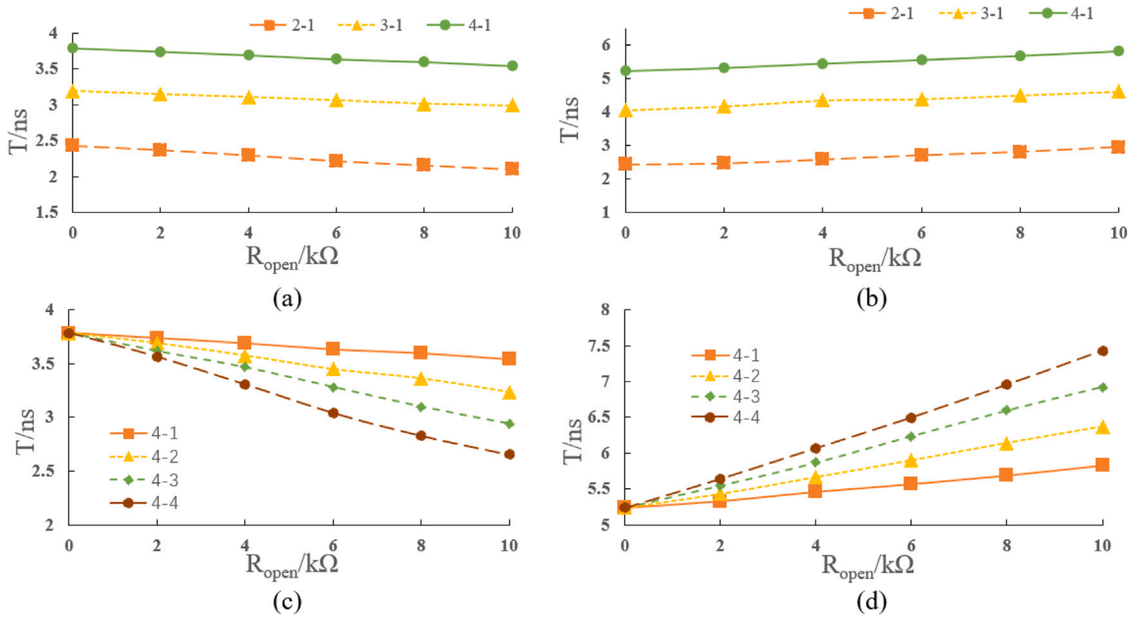

Fig. 10. Enter Oscillation period for different open faults. (a) Total number of TSVs in pre-bonding. (b) Total number of TSVs in post-bonding. (c) The number of faulty TSVs in pre-bonding. (d) Number of faulty TSVs in post-bonding.

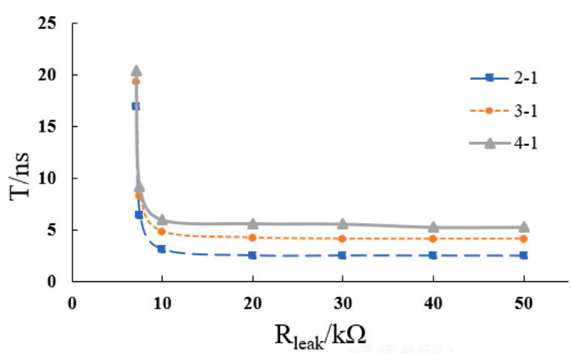

Fig. 11. Oscillation period for different leakage faults.

treatment can also be found in existing TSV test studies. [5] points out in pre-bond TSV testing that when the open fault is located near the bottom of the TSV, the effective capacitance observed from the test terminal changes only slightly, and therefore the test response is close to the fault-free reference state. As the fault location gradually moves away from the bottom, the equivalent capacitance variation becomes more significant, and the test response gradually deviates from the fault-free state. Therefore, an open fault near the bottom of the TSV can be regarded as a boundary case that is difficult to distinguish using this type of capacitance/delay-based test method.

Since the electrical model of a leakage defect is similar in both the pre-bond and post-bond stages, the experimental results remain consistent. Fig. 12(b) shows the results for leakage defects occurring at different locations. Because the maximum leakage resistance used in our experiments is 50 kΩ, we set the leakage defect to 30 kΩ for evaluation. In the figure, the horizontal axis represents the oscillation period, and the vertical axis denotes the fault location. When h=1/3, it indicates that the defect is located at one-third of the total TSV height. “4–1 (post-bond)” represents the oscillation period when one out of four TSVs contains a leakage defect during post-bond testing, while “post-bond fault-free period” represents the oscillation period when all four TSVs are fault-free. The results show that the position of the leakage defect has no impact on the oscillation period, and the period is always larger than that of the post-bond fault-free TSV. Therefore, our method can reliably detect leakage defects regardless of their location within the TSV.

Fig. 13 illustrates the relationship between the composite defect and the oscillation period. Since the maximum leakage resistance observed in our experiments is 50 kΩ, we set the leakage component of the composite defect to 30 kΩ for analysis. The horizontal axis represents the magnitude of the open defect, while the vertical axis represents the corresponding oscillation period. “4–1 (Pre-bond)” and “4–1 (Post-bond)” represent the oscillation period when one out of four TSVs contains a composite defect during pre-bond or post-bond testing. “Pre-bond fault-free period” and “Post-bond fault-free period” represent the oscillation period when all four TSVs are fault-free. In the pre-bond stage, when only the 30 kΩ leakage defect is present, the oscillation period is slightly larger than that of the fault-free case. As the open resistance increases, the period gradually decreases and eventually overlaps with the fault-free value, which may lead to a missed detection. In the post-bond stage, both leakage and open defects increase the oscillation period. Consequently, the period remains consistently higher than that of the fault-free case, enabling reliable detection of this type of composite fault.

Existing RO-based TSV test studies have shown that the absolute oscillation frequency or period of a ring oscillator is sensitive to random process variations and test-environment changes [17]. Therefore, such methods usually do not rely only on a single absolute oscillation period. Instead, reference paths or differential measurements are often adopted to reduce the influence of PVT variations on the test results [18]. For example, by comparing the oscillation periods when the TSV is bypassed and when the TSV is inserted into the test path, the common PVT-sensitive delay components introduced by logic gates, interconnects, and test structures can be partially canceled. As a result, the relative delay variation introduced by the TSV can be highlighted. For the proposed reconfigurable ring oscillator structure, a similar differential reference test can also be supported by introducing switches at both ends of the TSV, which can further improve the test stability under PVT variations. However, this approach requires additional switches and control logic, leading to extra area overhead and structural redundancy. The objective of this work is to reuse existing logic gates to construct a reconfigurable ring oscillator for testing a group of TSVs, and to use LFSR-generated control signals to reduce additional test circuitry, test

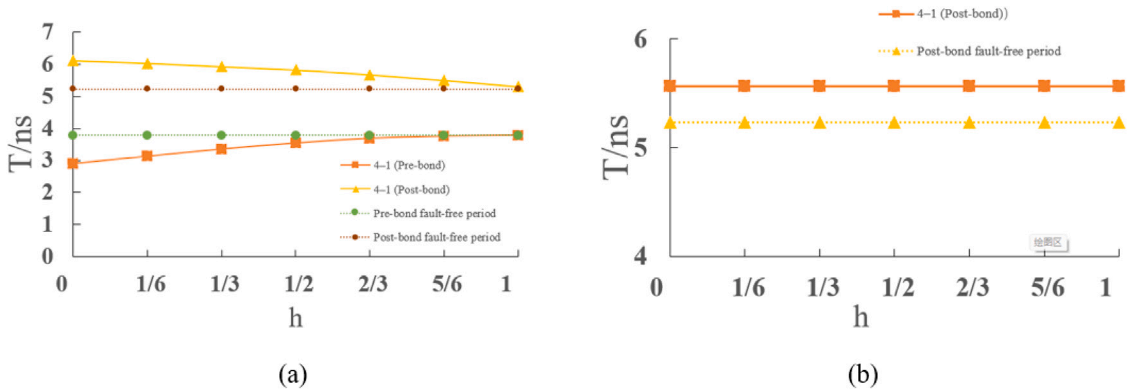

Fig. 12. Oscillation periods of open and leakage defects at different locations.

Table 1
Experimental circuit parameters.

Circuit	Reference circuit included	Inverters	Number of transmission gates	Number of triggers	Number of test data bits
3DIC-1	s1196 s1238	221	816	36	768
3DIC-2	s420 s5378	1853	1144	195	12 840
3DIC-3	s13207 s15850	5378	2573	638	408 100

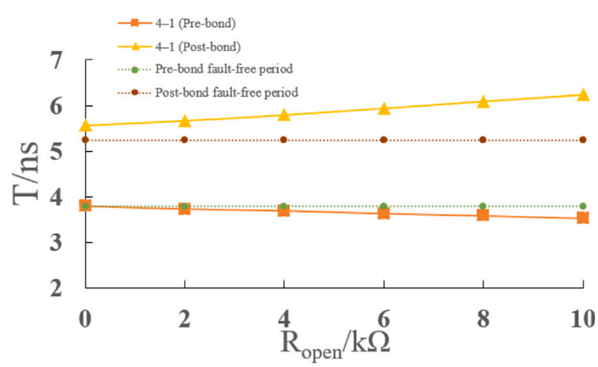

Fig. 13. The oscillation periods of different composite faults in the pre-bond and post-bond stages.

time, and test-data storage. Therefore, the proposed structure represents a tradeoff among area overhead, structural reusability, multi-TSV test efficiency, and test accuracy.

After confirming the effectiveness of this architecture in detecting TSV faults, it is imperative to validate its efficacy in circuit testing further. The experiment utilizes a three-dimensional chip with a three-layer structure as a case study. It integrates certain circuits from the ISCAS '89 reference circuit to form the experimental circuit, with specific parameters detailed in Table 1. The first column in Table 1 enumerates the names of experimental circuits at each layer. 3DIC-1, 3DIC-2, and 3DIC-3 simulate a three-layer 3D-SIC, with the scale of circuits increasing progressively across layers. The second column presents the reference circuits included in each layer of experimental circuits; each layer comprises two ISCAS '89 reference circuits. The third through sixth columns indicate the number of inverters, logic gates, flip-flops, and total test data bits contained within each layer of the circuit.

We compared performance against Refs. [15,19] on test data storage capacity, test data compression rate, and area cost with the 45 nm standard library in Table 2. It should be noted that the area overhead compared for all methods in Table 2 only includes the core test structures that are directly inserted into or reconfigured in the TSV test path. The counter and isolation circuits are not included in the per-TSV core area evaluation, because these modules do not directly constitute the oscillation path of each TSV and can usually be shared by multiple TSV test units. Therefore, Table 2 compares the area of the core test circuitry of each method. It is assumed that for each layer, the number of TSVs corresponds to the longest scan chain length for that particular layer. Ying et al. [19] introduced a parallel operation scheme using Dual-LFSR, which employed a mask LFSR to invert test data before merging and compressing it with the main LFSR's test data to reduce storage requirements. [15] proposed an RTS (reconfigurable LFSR low-power structure) that repurposes pre-bond test structures post-bond by reconstructing them into lower power consumption testing structures while allowing dynamic adjustment of power consumption during testing within low-power modules on-chip to minimize testing costs.

For each scheme, the test data storage capacity and test data compression ratio on different experimental circuits are given, where the compression ratio of test data is the compression ratio of the new test data to the original test data. The test data storage capacity of this paper was reduced by 53.41% compared to the double LFSR structure [19] and 24.39% compared to the RTS structure [15]. Our method RRO has an extraordinary enhancement in test data storage capacity and area overhead without losing fault coverage. In terms of area overhead, compared with the Dual-LFSR structure, the area overhead was reduced by 42.33% on average. Compared with the RTS structure, which was reduced by 19.87%. The reconfigurable structure has conspicuous advantages in overall area overhead. Compared with the test data storage without RRO, the increase in one used to configure RRO is only 0.39% on average, and the test data compression rate remains unaltered, which has superiorities over the other two methods.

We compare the proposed method with the five state-of-the-art methods [6,9,20,21], and [22] in terms of test stage, detectable fault types, technology node, and area overhead. The results are summarized in Table 3, where the reported area overhead corresponds to the case of testing 100 TSVs. For TSV testing, the aforementioned methods can only be applied either in pre-bonding or post-bonding, and among them, only the pulse-shrinking method in [6] is capable of detecting

Table 2
Comparison results with other methods.

Circuit	Dual-LFSR [19]			RTS [15]			RCT			RCT with RRO		
	Test data storage capacity	Test data compression rate	Area overhead	Test data storage capacity	Test data compression rate	Area overhead	Test data storage capacity	Test data compression rate	Area overhead	Test data storage capacity	Test data compression rate	Area overhead
3DIC-1	624	18.75%	231.42	420	45.31%	131.40	150	80.47%	135.41	164	80.29%	135.41
3DIC-2	2280	82.24%	319.20	2720	78.82%	333.294	1641	87.22%	232.34	1710	87.11%	232.34
3DIC-3	51887	87.29%	734.16	30616	92.95%	459.91	23734	94.18%	373.14	23754	95.06%	373.14
AVG	18263	87.01%	428.26	11252	92.00%	308.2	8508	93.95%	246.96	8542	93.95%	246.96
Comparison	−53.41%	+6.94%	−42.33%	−24.39%	+2.11%	−19.87%	\	\	\	+0.39%	\	\

Table 3
Comparison of test methods.

Test methods	Test stage	Detectable fault types	Technology	Area Overhead (μm^2)
Method in [6]	Pre-bond	Leakage, Resistive Open, Composite Faults	45 nm	242.858
Method in [9]	Post-bond	Resistive Open	65 nm	490
Method in [20]	Pre-bond	Leakage, Resistive Open	45 nm	673.192
Method in [21]	Pre-bond	Resistive Open, Leakage	90 nm	374
Method in [22]	Pre-bond	Resistive Open, Leakage	45 nm	778.2
Proposed Method	**Pre-bond, Post-bond**	**Resistive Open, Leakage, Composite Faults**	**45 nm**	**20**

resistive-open, leakage, and composite faults simultaneously. In contrast, the proposed method supports both pre-bond and mid/post-bond TSV testing. By configuring logic gates to form RO loops at different test stages, our approach creates multiple detection paths, enabling the detection of all three fault types. Furthermore, the control signals are generated by the existing RLFSR modules in the circuit, requiring no additional area overhead. Only one redundant TSV is added to the CUT scan chain to configure the logic gates and construct the RRO. The area overhead of a single TSV is 20 μm^2 [23], which is smaller than that of existing methods. The proposed method offers an advantage in terms of area efficiency in contrast to the compared methods.

5. Conclusion

This article proposes a reconfigurable RO structure based on RLFSR, which can effectively detect faults of the TSV and chip circuits of a three-dimensional chip, thereby ensuring the chip yield. The proposed structure utilizes a reconfigurable RO to detect TSV. It determines whether the TSV has an open fault or a leakage fault according to the size of the oscillation period. It detects faulty TSVs in pre-bonding or in the process of mid-bond testing and post-bond testing. The RO reforms a new one to test TSV by multiplexing the RO in pre-bonding. The reconfigurable LFSR in the proposed architecture generates test vectors for different stages of 3D chip bonding, and simultaneously provides the control signals required to configure the RO-based paths for TSV testing.

The experimental results demonstrate that the proposed testing method can effectively validate TSV functionality across different chip bonding stages. In addition, compared with existing approaches, the proposed method exhibits clear advantages in terms of test cost and hardware overhead. Specifically, relative to the methods in [15,19], it reduces the required test-data storage by 53.41% and 24.39%, respectively, and decreases the area overhead by 42.33% and 19.87%, respectively. For TSV testing, the proposed method can be applied in both pre-bond and post-bond stage, and is capable of detecting open faults, leakage faults, and composite faults. Moreover, the area overhead of the proposed method is significantly lower, as it requires only one additional TSV, with an area of approximately 20 μm^2. These results indicate that the proposed approach provides a more efficient TSV testing solution, while maintaining compatibility with multi-stage TSV testing and ensuring low area overhead.

CRediT authorship contribution statement

Tian Chen: Conceptualization, Methodology, Writing – review & editing, Funding acquisition. **Daping Wu:** Methodology, Formal analysis, Writing – original draft, Writing – review & editing. **Yunfei Zhang:** Software, Validation. **Jun Liu:** Conceptualization, Writing – review & editing. **Weikun Chen:** Software, Validation. **Xiaohui Yuan:** Writing – original draft, Writing – review. **Yingchun Lu:** Investigation. **Huaguo Liang:** Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the National Natural Science Foundation of China under Grant 62174048 and Grant 62027815.

Data availability

Data will be made available on request.

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